

Table 4: Transfer learning performance comparison.

Model	Volume to Band gap (eV)		Band gap to Volume($\text{\AA}^3/\text{cell}$)	
	Validation set ↓	Test set ↓	Validation set ↓	Test set ↓
Structure-based models				
CGCNN (transfer)	0.300	0.295	187.473	182.997
MEGNet (transfer)	1.461	1.472	190.013	195.664
ALIGNN (transfer)	0.321	0.322	137.471	136.164
Text-based models				
MatBERT (zero-shot)	1.175	1.191	412.891	422.487
MatBERT (transfer)	0.259	0.266	50.530	54.289
LLM-Prop (zero-shot)	1.288	1.103	482.922	485.583
LLM-Prop (transfer)	0.236	0.244	47.837	50.753

Table 5: LLM-Prop performance (AUC ROC score) comparison with baselines on classifying whether the band gap value is direct or indirect.

Model	#Parameters	Is-gap-direct	
		Validation set ↑	Test set ↑
Structure-based models			
CGCNN	-	0.797	0.830
ALIGNN	-	0.814	0.678
Text-based models			
MatBERT (zero-shot)	109.5M	0.500	0.535
MatBERT		0.722	0.728
LLM-Prop (zero-shot-512 tokens)		0.494	0.484
LLM-Prop (512 tokens)	37M	0.839	0.836
LLM-Prop (zero-shot)		0.503	0.500
LLM-Prop		0.856	0.857

yields better or comparable performance, including for zero-shot prediction, on all tasks (2, 3, and 5). When we finetune LLM-Prop on the maximum input sequence length it can process (888 tokens), it outperforms MatBERT by a large margin despite having $3\times$ fewer hyperparameters. LLM-Prop improves upon MatBERT by approximately 6% and 3% on band gap prediction on the validation and test sets, respectively (see Table 2), 15% and 16% on volume prediction (see Table 3), and 19% and 18% on predicting whether the band gap is direct or indirect (see Table 5). Although MatBERT was pretrained on in-domain data, LLM-Prop shows stronger transfer learning ability when transferring between models finetuned on volume and band gap (see Table 4). The MatBERT results show that naively finetuning pretrained LLMs (even though they have been pretrained on domain specific data) is insufficient for accurate crystal property prediction and additional strategies such as those used in this paper are needed to achieve better performance.

How much data does LLM-Prop need to achieve SOTA results? We finetune LLM-Prop with randomly sampled data from the training set with the data size